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Online Videos Week 6

Topic 2: Torsion, Bending & Beam Elements

KB5034 Mechanics and Finite Element Analysis

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Learning Outcome 2.6

By the end of this lesson, students will be able to…

Compare the order of shape functions for beam and truss elements.

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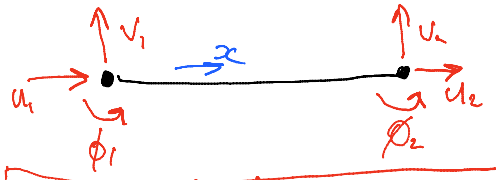
Method of Virtual Work for Beam Elements

Step 2. Define deflection as a function of nodal degrees of freedom

$$v = f(\underline{u})$$

v_1, ϕ_1, v_2, ϕ_2

4 unknowns



Shape functions give

Shape functions interpolate

Deflection assumed to be cubic parabola

at $x=0$ $\quad v(0) = v_1 \quad \frac{dv}{dx}(0) = \phi_1$

at $x=L$ $\quad v(L) = v_2 \quad \frac{dv}{dx}(L) = \phi_2$

4 equations, 4 unknowns

Solve for a, b, c and d in terms of v_1, v_2, ϕ_1 and ϕ_2

$a =$ $\qquad c =$

$b =$ $\qquad d =$

$$v(x) = a + bx + cx^2 + dx^3$$

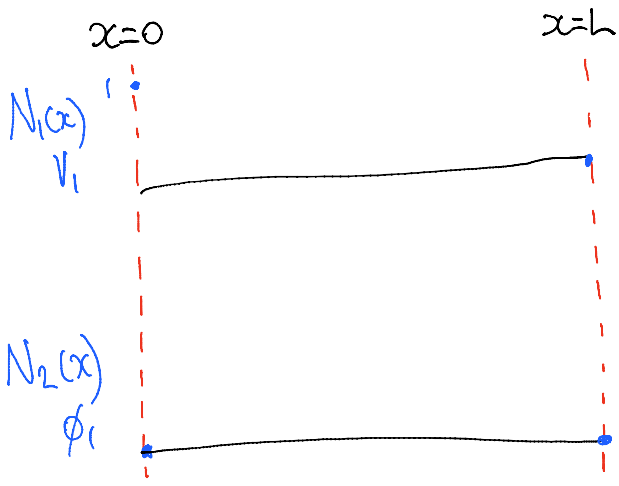
$v(x) =$

$$v(x) = N_1(x)\, v_1 + N_2(x)\, \phi_1 + N_3(x)\, v_2 + N_4(x)\, \phi_2$$

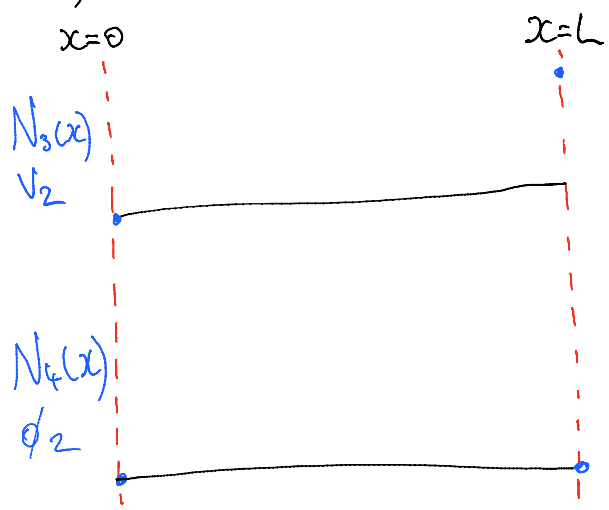
$N_1(x) =$ $\qquad N_3(x) =$

$N_2(x) =$ $\qquad N_4(x) =$

$N_2(x)$ =



$N_4(x)$ =



Step 3. Derive strain-displacement matrix $\underline{B}$

$$\underline{\varepsilon} = f(\underline{u}) \rightarrow \underline{\varepsilon} = \underline{B}\,\underline{u}$$

$$V(x) = N_1(x)V_1 + N_2(x)\phi_1 + N_3(x)V_2 + N_4(x)\phi_2$$

$$\varepsilon = y\frac{d^2V(x)}{dx^2}$$

$$V''(x) = N_1''(x)V_1 + N_2''(x)\phi_1 + N_3''(x)V_2 + N_4''(x)\phi_2$$

$N_1(x) = 1 - 3\frac{x^2}{L^2} + 2\frac{x^3}{L^3}$ $\quad N_2'' =$ $\quad N_4'' =$

$N_1'(x) =$ $\quad N_3'' =$

$N_1''(x) =$

$$\underline{\varepsilon} = y\begin{bmatrix} N_1'' & N_2'' & N_3'' & N_4'' \end{bmatrix}\begin{bmatrix} V_1 \\ \phi_1 \\ V_2 \\ \phi_2 \end{bmatrix}$$

Step 4. Use method of virtual work to determine $\underline{k}$

$$\int_V \underline{B}^T\underline{E}\,\underline{B}\,\underline{u}\,dV = \underline{f}$$

$\underline{B}^T = y\begin{bmatrix} N_1'' \\ N_2'' \\ \vdots \end{bmatrix}$ $\quad \begin{bmatrix} N_1'' & N_2'' & N_3'' & N_4'' \end{bmatrix}$

$$\underline{k}'_e = \int_V \underline{B}^T \underline{E} \underline{B}\, dV \qquad \underline{B} = y[N_1'' \; N_2'' \; N_3'' \; N_4''] \qquad \underline{B}^T = y\begin{bmatrix} N_1'' \\ N_2'' \\ N_3'' \\ N_4'' \end{bmatrix}$$

E is constant throughout element

$\underline{k}'_e =$

$I =$

$$\underline{k}'_e = EI \int_x \begin{bmatrix} N_1''N_1'' & N_1''N_2'' & N_1''N_3'' & N_1''N_4'' \\ N_2''N_1'' & N_2''N_2'' & N_2''N_3'' & N_2''N_4'' \\ N_3''N_1'' & N_3''N_2'' & N_3''N_3'' & N_3''N_4'' \\ N_4''N_1'' & N_4''N_2'' & N_4''N_3'' & N_4''N_4'' \end{bmatrix} dx$$

$$N_1'' = \frac{-6}{L^2} + \frac{12x}{L^3}$$

$$k'_{11} = EI \int_{x=0}^{x=L} N_1''(x)\, N_1''(x)\, dx = EI \int_0^L \left(\frac{-6}{L^2} + \frac{12x}{L^3}\right)^2 dx$$

$k'_{11} =$

$k'_{11} =$

$k'_{11} =$

$k'_{11} =$

$$k'_{11} = \frac{12EI}{L^3}$$

$$\underline{k}'_e = EI \int_x \begin{bmatrix} N_1''N_1'' & N_1''N_2'' & N_1''N_3'' & N_1''N_4'' \\ N_2''N_1'' & N_2''N_2'' & N_2''N_3'' & N_2''N_4'' \\ N_3''N_1'' & N_3''N_2'' & N_3''N_3'' & N_3''N_4'' \\ N_4''N_1'' & N_4''N_2'' & N_4''N_3'' & N_4''N_4'' \end{bmatrix} dx$$

$$\begin{bmatrix} f_{1y} \\ \vdots \end{bmatrix} \begin{bmatrix} 12\frac{EI}{L^3} & 6EI/L^2 & -12EI/L^3 & 6EI/L^2 \\ \vdots & 4EI & -6EI/L^2 & 2EI/L \end{bmatrix} \begin{bmatrix} v_1 \\ \phi_1 \end{bmatrix}$$

$$\begin{bmatrix} f_{1y} \\ M_1 \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} 12\frac{EI}{L^3} & \frac{6EI}{L^2} & \frac{-12EI}{L^3} & \frac{6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{4EI}{L} & \frac{-6EI}{L^2} & \frac{2EI}{L} \\ \frac{-12EI}{L^3} & \frac{-6EI}{L^2} & \frac{12EI}{L^3} & \frac{-6EI}{L^2} \\ \frac{6EI}{L^2} & \frac{2EI}{L} & \frac{-6EI}{L^2} & \frac{4EI}{L} \end{bmatrix} \begin{bmatrix} \phi_1 \\ V_2 \\ \phi_2 \end{bmatrix}$$

$$\begin{bmatrix} f_{1x} \\ f_{1y} \\ M_1 \\ f_{2x} \\ f_{2y} \\ M_2 \end{bmatrix} = \begin{bmatrix} \frac{EA}{\ell} & 0 & 0 & -\frac{EA}{\ell} & 0 & 0 \\ 0 & \frac{12EI}{\ell_3} & \frac{6EI}{\ell^2} & 0 & \frac{-12EI}{\ell^3} & \frac{6EI}{\ell^2} \\ 0 & \frac{6EI}{\ell^2} & \frac{4EI}{\ell} & 0 & \frac{-6EI}{\ell^2} & \frac{2EI}{\ell} \\ -\frac{EA}{\ell} & 0 & 0 & \frac{EA}{\ell} & 0 & 0 \\ 0 & \frac{-12EI}{\ell^3} & \frac{-6EI}{\ell^2} & 0 & \frac{12EI}{\ell^3} & \frac{-6EI}{\ell^2} \\ 0 & \frac{6EI}{\ell^2} & \frac{2EI}{\ell} & 0 & \frac{-6EI}{\ell^2} & \frac{4EI}{\ell} \end{bmatrix} \begin{bmatrix} u_1 \\ v_1 \\ \phi_1 \\ u_2 \\ v_2 \\ \phi_2 \end{bmatrix}$$